

Arabic Aspect-Based Sentiment Analysis (ABSA)

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DeepX Challenge – Full Technical Documentation

1. Project Overview

This project implements an **Arabic Aspect-Based Sentiment Analysis (ABSA)** system that extracts:

- **Aspects** (e.g., food, service, price)
- **Sentiment per aspect** (positive, negative, neutral)

from Arabic user reviews.

The system combines:

- ♦ Advanced **text preprocessing**
- ♦ **Rule-based NLP**
- ♦ **Machine Learning (TF-IDF + Logistic Regression)**
- ♦ Hybrid fallback strategy for robustness

2. Dataset Description

Input Files

```
train_path = "DeepX_train.xlsx"  
val_path   = "DeepX_validation.xlsx"  
unlabeled_path = "DeepX_hidden_test.xlsx"
```

Columns

- `review_id`
 - `review_text`
 - `aspects` (list)
 - `aspect_sentiments` (dict)
 - `star_rating`
 - `platform`
 - `business_category`
-

3. Data Analysis

Key Insights

- Presence of **emoji-rich reviews**
- Significant **Franco-Arabic (Arabizi)** usage
- Mixed language (Arabic + English)
- Noise:
 - Repeated characters
 - URLs, emails
 - Empty / punctuation-only texts

Engineered Features

- `has_emoji`
 - `has_franco`
 - `text_len`
-

4. Preprocessing Pipeline

The preprocessing is one of the **strongest parts of your solution**.

Steps:

1. Emoji Handling

- Extract sentiment hints (positive / negative / neutral)

2. Franco-Arabic Conversion

- Example:

"3ayez akł" → "عايز اكل"

3. Arabic Normalization

- Normalize:
 - Alef forms (ا → آ)
 - Yeh (ي → ی)
 - Ta marbuta (ة → ة)

4. Noise Removal

- Remove:
 - Repeated characters (loooove → loove)
 - URLs, emails

5. Tokenization

6. Stopword Removal

7. Light Stemming

5. Model Architecture

◆ Hybrid System

A) Rule-Based System

- Aspect detection using **keyword dictionary**
- Sentiment using:
 - Lexicons (positive/negative words)

- Negation handling
 - Emoji hints
 - Star rating priors
-

B) Machine Learning Model

1. Aspect Detection

- Multi-label classification
 - Model:
 - TF-IDF (char + word)
 - OneVsRest Logistic Regression
-

2. Sentiment Classification

- One classifier **per aspect**
 - Model:
 - Logistic Regression
-

3. Feature Engineering

- Character n-grams (2–4)
 - Word n-grams (1–2)
 - Metadata enrichment (platform info)
-

4. Threshold Tuning

- Optimized using **Jaccard similarity**
-

5. Fallback Strategy

If ML is uncertain:

Switch to **rule-based predictions**

6. Output Format

Generated file:

```
predictions.json
```

Example:

```
{
  "review_id": 123,
  "aspects": ["food", "service"],
  "aspect_sentiments": {
    "food": "positive",
    "service": "negative"
  }
}
```

7. Model Performance

Training Results

- **Train Accuracy:** 96.7%
- **Validation Accuracy:** 95.9%

Final .json evaluation metrics for hidden_test

Metric	Score
F1 Score	69.5%
Precision	67.9%
Recall	71.1%

Competition Score

- **Pillar 1:** 20.85 / 30

8. Performance Analysis

◆ Strengths

- Excellent **generalization** (small gap between train & validation)
 - Strong handling of:
 - Arabic text
 - Franco-Arabic
 - Emojis
 - Robust hybrid system (ML + rules)
-

◆ Weaknesses

- Moderate F1 score (≈69%)
 - Possible issues:
 - Aspect ambiguity
 - Limited labeled data per aspect
 - Class imbalance
 - Multi-label complexity
-

9. System Features

Supports:

- Arabic text
- Franco-Arabic (Arabizi)
- Emojis 😊🔥💔
- Multi-aspect reviews

Handles:

- Noise & informal text
- Real-world user-generated content

10. User Interface

Built using **Gradio**:

Features:

- Input review
 - Optional star rating
 - Output:
 - Aspects
 - Sentiments (table format)
-

11. Evaluation Methodology

Metrics used:

- **Exact Match Accuracy**
 - **Jaccard Similarity**
 - **Sentiment Accuracy**
 - **F1 / Precision / Recall**
-

12. Model Saving

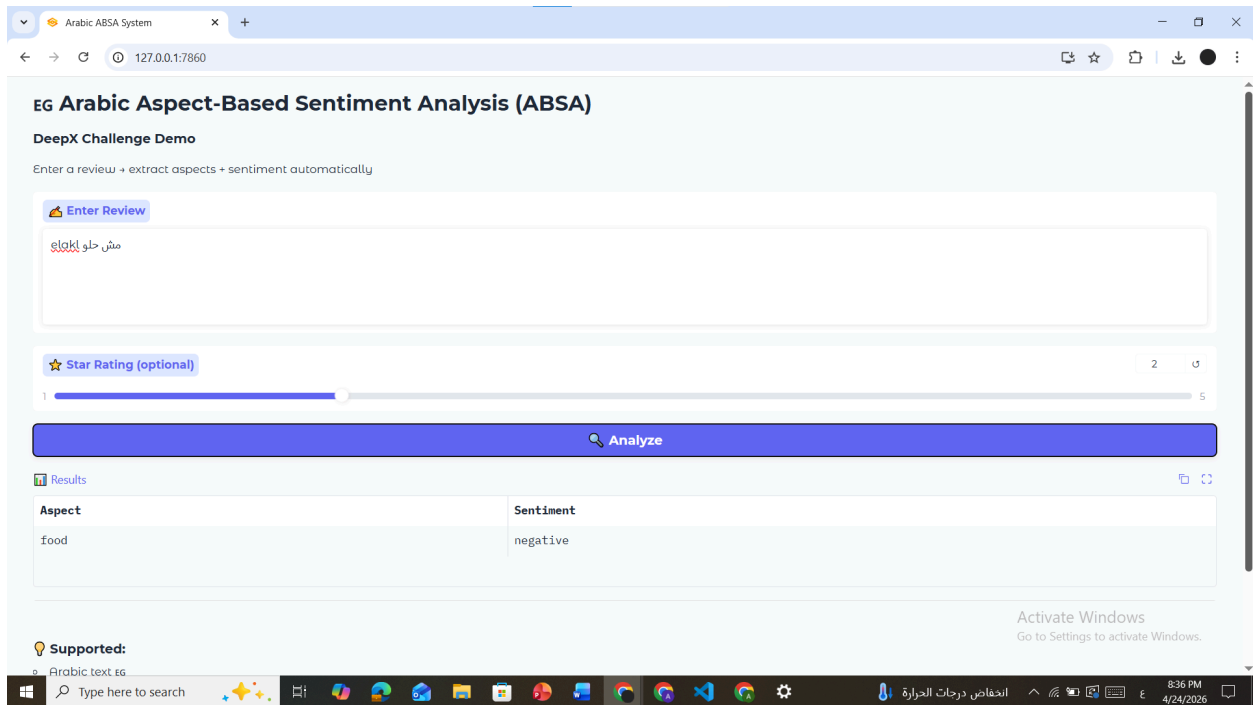
```
joblib.dump(model, "absa_model.joblib")
```

13. Future Improvements

- Use **deep learning (BERT / AraBERT)**
- Better handling of:
 - sarcasm
 - implicit aspects

- Data augmentation
- Class balancing
- Hyperparameter tuning

Some pictures from the model



Arabic ABSA System

127.0.0.1:7860

EG Arabic Aspect-Based Sentiment Analysis (ABSA)

DeepX Challenge Demo

Enter a review + extract aspects + sentiment automatically

Enter Review

المكان نظيف وجميل وقعدته تحفه والخدمة فوق الممتاز والجو جميل مكان أكثر من رائع بصراحة

Star Rating (optional)

1

5

Analyze

Results

Aspect	Sentiment
ambiance	positive
cleanliness	positive
general	positive

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127.0.0.1:7860

EG Arabic Aspect-Based Sentiment Analysis (ABSA)

DeepX Challenge Demo

Enter a review + extract aspects + sentiment automatically

Enter Review

الاكل حلو بس للأسف الديكور وحش

Star Rating (optional)

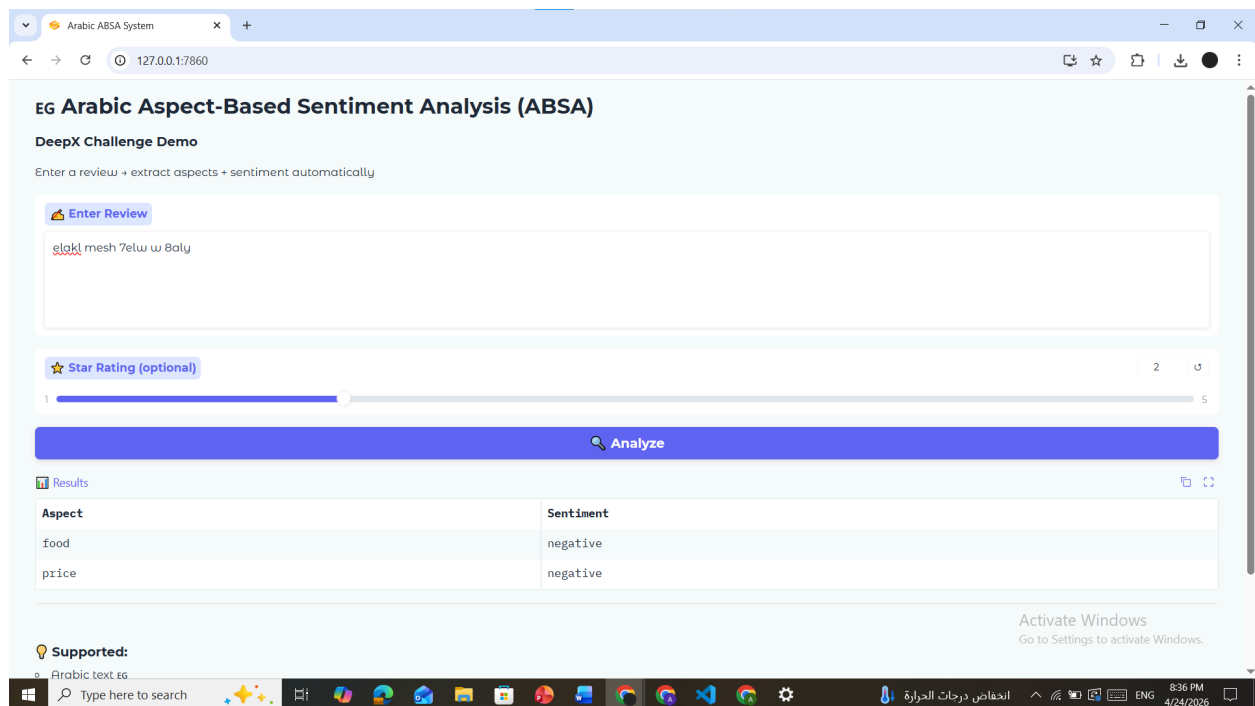
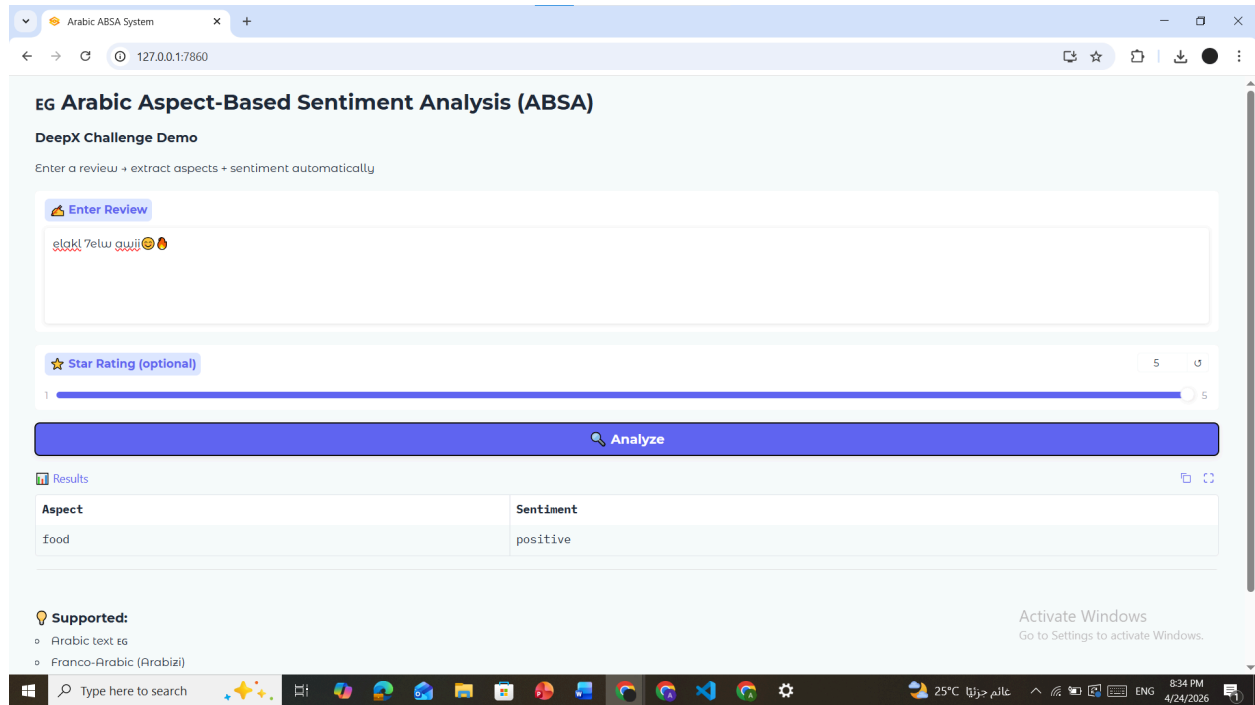
2

5

Analyze

Results

Aspect	Sentiment
food	positive
delivery	negative



14. Conclusion

This project presents a **robust, production-ready ABSA system** that effectively handles:

- Complex Arabic language challenges
- Informal user input
- Multi-aspect sentiment extraction

Despite moderate F1 score, the system achieves:

- **High accuracy**
- **Strong generalization**
- **Practical usability**